MILITARY SPECIFICATION

ADHESIVE, POLYURETHANE

This specification is approved for use within the Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of two-part polyurethane adhesive.

1.2 Military part number. The military part number shall consist of the letter B, the basic number of this specification, and dash letters denoting container size.

Example: B 47315 - PT

- PT — Container size (see table I)
- 47315 — Basic specification number
- B — Letter B denoting bulk material

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN:AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL, 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 8040

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

TABLE I. Container size.

Container 1/	Dash identification designator
Ounce	OZ
Pint	PT
Quart	QT
Gallon	GL

1/ Metric equivalents shall be acceptable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

MMM-A-132 Adhesives, Heat Resistant, Airframe Structural, Metal-to-Metal

PPP-C-2020 Chemicals, Liquid, Dry, and Paste, Packaging of

STANDARDS

FEDERAL

FED-STD-175 Adhesives, Methods of Testing

FED-STD-406 Plastics, Methods of Testing

MILITARY

MIL-STD-129 Marking for Shipment and Storage

MIL-STD-130 Identification Marking of U.S. Military Property

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

MIL-STD-810 Environmental Test Methods and Engineering Guidelines

MIL-STD-45662 Calibration Systems Requirements

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D150 Insulating Materials, Solid Electrical, AC Loss Characteristics and Permittivity (Dielectric Constant)

ASTM D792 Specific Gravity and Density of Plastics By Displacement

ASTM D2240 Rubber, Property-Durometer Hardness

ASTM D3951 Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified in the contract or procurement document, a first article sample shall be provided which shall meet the requirements of this specification (see 4.4.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

3.2 Properties.

3.2.1 Uncured properties. The adhesive, as received, shall be polyurethane polyether base reversion resistant premixed and frozen.

3.2.1.1 Specific gravity. The specific gravity shall be 1.25 ± 0.05 at temperature of 25° ± 2°C (77° ± 4°F), (see 4.6.3.1).

3.2.1.2 Working life. The usable life of the mixed compound, exposed in an open container at 25° ± 2°C (77° ± 4°F), shall be not less than 45 minutes (see 4.6.3.2).

3.2.2 Cured properties. The compound shall exhibit the following properties after curing for 3 hours, minimum, at a temperature of 71°C + 3°C (160° + 4°F) and allowed to cool to a temperature of 25° ± 2°C (77° ± 4°F), (see 4.6.4).

3.2.2.1 Overlap shear strength. (See 4.6.4.1).

Temperature	Shear Strength, Psi
25° ± 2°C (77° ± 4°F)	2000 ± 500
-54° ± 3°C (-65° ± 4°F)	4000 ± 1000
68.5° ± 3°C (155° ± 4°F)	1000 ± 250

3.2.2.2 Hardness, Shore A. The Shore A hardness shall be 90 ± 5 (see 4.6.4.2).

3.2.2.3 Dielectric constant. The dielectric constant shall be 6.6, maximum at 1 kilohertz (KHz), (see 4.6.4.3).

3.2.2.4 Volume resistivity. The volume resistivity shall be 9×10^9 ohm/centimeter, minimum at 25°C (77°F), (see 4.6.4.4).

3.2.2.5 Copper corrosion. The compound shall not exhibit corrosive effects from copper (see 4.6.4.5).

3.2.2.6 Fungus resistance. The compound shall not exhibit effects to fungus nutrients (see 4.6.4.6).

3.2.2.7 Hydrolytic stability. The compound shall not exhibit a reversion characteristic. The loss of hardness shall be not greater than 10 (see 4.6.4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

3.3 Storage life. The premixed and frozen compound, when stored in sealed containers, shall meet the requirements of 3.2 after a storage period of 30 days minimum, at -40°C (-40°F) or colder (see 4.7).

3.4 Marking for identification. Marking for identification shall be in accordance with MIL-STD-130 and shall contain at least the following information: (See 4.6.1).

a. Adhesive polyurethane.

b. Military part number.

c. Quantity (net weight per container).

d. Vendor's name or trademark.

e. Date of manufacture, lot or batch (month and year).

f. Any pertinent usage, warning, or storage instructions.

3.5 Workmanship. The adhesive compound shall be prepared in a workmanlike manner and shall be free of apparent foreign particles (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsiblity for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under ambient conditions.

4.3.1 Inspection equipment and facilities. Inspection equipment and facilities shall be of adequate quality and quantity to permit performance of the required inspections.

4.3.2 Inspection equipment accuracy. Equipment used to measure test conditions and test parameters shall be calibrated in accordance with MIL-STD-45662. Where the state of the art permits, basic precision equipment capable of measuring or testing to an accuracy representing one-tenth of the prescribed tolerance of the inspection equipment shall be used.

4.4 First article inspection.

4.4.1 First article test sample. A first article sample of not less than one quart shall be provided. The first article sample shall be fabricated using the same processes, facilities, and materials as are proposed for production.

4.4.2 First article tests. The first article tests shall be as specified in table II.

4.4.3 Testing sequence. Testing sequence for first article inspection shall be in the order as listed in table II.

TABLE II. First article inspection.

Examination and test	Requirement paragraph	Method paragraph
Specific gravity	3.2.1.1	4.6.3.1
Working life	3.2.1.2	4.6.3.2
Overlap shear strength	3.2.2.1	4.6.4.1
Hardness, Shore A	3.2.2.2	4.6.4.2
Dielectric constant	3.2.2.3	4.6.4.3
Volume resistivity	3.2.2.4	4.6.4.4
Copper corrosion	3.2.2.5	4.6.4.5
Fungus resistance	3.2.2.6	4.6.4.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

TABLE II. First article inspection - Continued.

Examination and test	Requirement paragraph	Method paragraph
Hydrolytic stability	3.2.2.7	4.6.4.4
Storage life	3.3	4.7
Marking for identification	3.4	4.6.1
Workmanship	3.5	4.6.1

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections specified in table III.

TABLE III. Quality conformance inspection.

Examination and Test	Requirement paragraph	Method paragraph
Overlap shear strength	3.2.2.1	4.6.4.1
Hardness, Shore A	3.2.2.2	4.6.4.2
Marking for identification	3.4	4.6.1
Workmanship	3.5	4.6.1
Packaging	Section 5	4.8

4.5.1 Inspection lot. An inspection lot for quality conformance inspection shall consist of material manufactured under uniform conditions and offered for acceptance at one time.

4.5.2 Quality conformance sample. Quality conformance inspection shall be performed on sufficient quantities of compound taken from a single lot to make one pint of adhesive.

4.5.3 Test sequence. Testing sequence shall be in the order as listed in table III.

4.5.4 Nonconformance. Failure of a sample to meet one or more of the specified requirements for quality conformance inspection shall constitute a basis for rejection of the entire lot. Subsequent lots shall not be offered for acceptance until technical judgment dictates that the cause of failure has been analyzed and solved to the satisfaction of the procuring activity.

4.6 Inspection methods.

4.6.1 Visual inspections. Visual inspection shall verify compliance with the requirements of 3.4, 3.5, and section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

4.6.2 Test sample preparation. The test sample preparation shall be in accordance with the applicable ASTM method, or standard.

4.6.3 Uncured properties.

4.6.3.1 Specific gravity. Specific gravity shall be determined in accordance with ASTM D792, method A. Verify compliance with 3.2.1.1.

4.6.3.2 Working life. Working life shall be determined by thawing out the compound in accordance with manufacturer's recommendations and applying the conditions of 3.2.1.2. Verify compliance with 3.2.1.2.

4.6.4 Cured properties.

4.6.4.1 Overlap shear strength. Overlap shear strength shall be determined in accordance with MMM-A-132. Verify compliance with 3.2.2.1.

4.6.4.2 Hardness, Shore A. Shore A hardness shall be determined in accordance with ASTM D2240. Verify compliance with 3.2.2.2.

4.6.4.3 Dielectric constant. Dielectric constant shall be determined in accordance with ASTM D150. Verify compliance with 3.2.2.3.

4.6.4.4 Hydrolytic stability and volume resistivity. Vendor certification is acceptable in lieu of testing (see figure 1).

4.6.4.4.1 Specimens. Four hardness specimens, approximately 2 inches in diameter and 0.250 inch thick, shall be prepared. Six specimens, 4 inches in diameter and approximately 0.075 inch thick, shall be prepared for volume resistivity tests.

4.6.4.4.2 Conditioning. Two hardness specimens and three volume resistivity specimens shall be maintained for 120 days at each of the following conditions:

25°C (77°F) and 50 percent relative humidity

71°C (160°F) and 95 percent relative humidity

4.6.4.4.3 Procedure. This test may be conducted in a chamber instead of sealed apparatus. The specimens shall be tested for hardness and volume resistivity after 0, 28, 84 and 120 days. The 71°C specimens shall be held at 25°C and 50 percent relative humidity for 2 hours prior to testing. The instantaneous hardness shall be determined using a Shore A durometer, in accordance with ASTM D2240. Volume resistivity tests shall be conducted in

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

accordance with FED-STD-406, method 4041 using a Beckman Megohmmeter Model L-8 or equivalent instrument with a test voltage of 500 volts. Readings shall be made 1 minute after application of current. Lead foil electrodes shall be a disk 2 inches in diameter, centrally located on one face of the specimen. The guard electrode shall be a concentric ring of 2.281 inches inside diameter and with an outside diameter equal to that of the specimen. The unguarded electrode shall be a foil disk 4 inches in diameter applied to the opposite side of the specimen. The test current shall be introduced to the guarded electrode, the guard electrode, and the unguarded electrode by means of a brass disk 2 inches in diameter by 1 inch in thickness, a brass ring 2.313 inches inside diameter by 4 inches outside diameter by 0.125 inch thick, and a brass disk 4 inches in diameter, respectively (see figure 1).

4.6.4.4.4 Calculation. The average volume resistivity shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

Where:

r = volume resistivity in ohm-centimeters.

R = volume resistance in megohms.

A = $(r_0)^2$, effective area of guarded electrode in square centimeters (see figure 1).

t = Average thickness of specimens in centimeters.

4.6.4.5 Copper corrosion. The copper corrosion test shall be performed in accordance with FED-STD-175, method 4031-1. Verify compliance with 3.2.2.5.

4.6.4.6. Fungus resistance. Fungus resistance shall be determined in accordance with MIL-STD-810, method 508. Vendor certification is acceptable in lieu of testing. Verify compliance with 3.2.2.6.

4.7 Storage life. The compound shall meet the requirements of 3.3 when stored in unopened containers for 30 days, minimum, at -40°C (-40°F) or colder.

4.8 Inspection of packaging. Materials ready for shipment shall be examined to determine conformance to the requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Levels of packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The compound shall be packaged in accordance with the overseas packaging requirements of PPP-C-2020.

5.1.2 Level B. The requirements of 5.1.1 shall apply for level B packaging.

5.1.3 Level C. The compound shall be packaged in unit capacity containers as specified (see 6.2 and table I) and shall conform to requirements of ASTM D3951.

5.2 Levels of packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The compound shall be packed in accordance with the overseas packing requirements of PPP-C-2020.

5.2.2 Level B. The compound shall be packed in accordance with 5.2.1, except shipping containers shall be domestic type.

5.2.3 Level C. Materials packaged as specified (see 6.2) shall be packed in containers which will insure acceptance by common carrier at lowest rates and provide safe delivery at destination.

5.3 Marking. In addition to the requirements specified in 3.4, the containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The adhesive covered by this specification is intended to bond metals and glass to themselves and to each other.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following:

a. Number, title, and date of this specification.

b. Military part number.

c. Quantity required and quantity per unit package of base.

d. Whether first article sample is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

e. Certification of fungus resistance in lieu of testing (see 4.6.4.6).

f. Official levels of packaging and packing required (see 5.1 and 5.2).

g. Sampling plan if other than specified (see 4.5).

h. Certification of storage life, in lieu of testing, if permissible (see 3.3 and 4.7).

6.2.2 First article. When a first article inspection is required, the item will be tested and should be a sample selected from the first production lot. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Storage life extension. Material that has exceeded storage life may be used upon verification of application time and durometer hardness. Acceptable test results shall constitute storage life extension of six months from date of test.

6.4 Subject term (key word) listing.

Adhesive
Polyurethane

Custodian:
Army - MI

Preparing activity:
Army - MI

Project No. 8040-A127

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

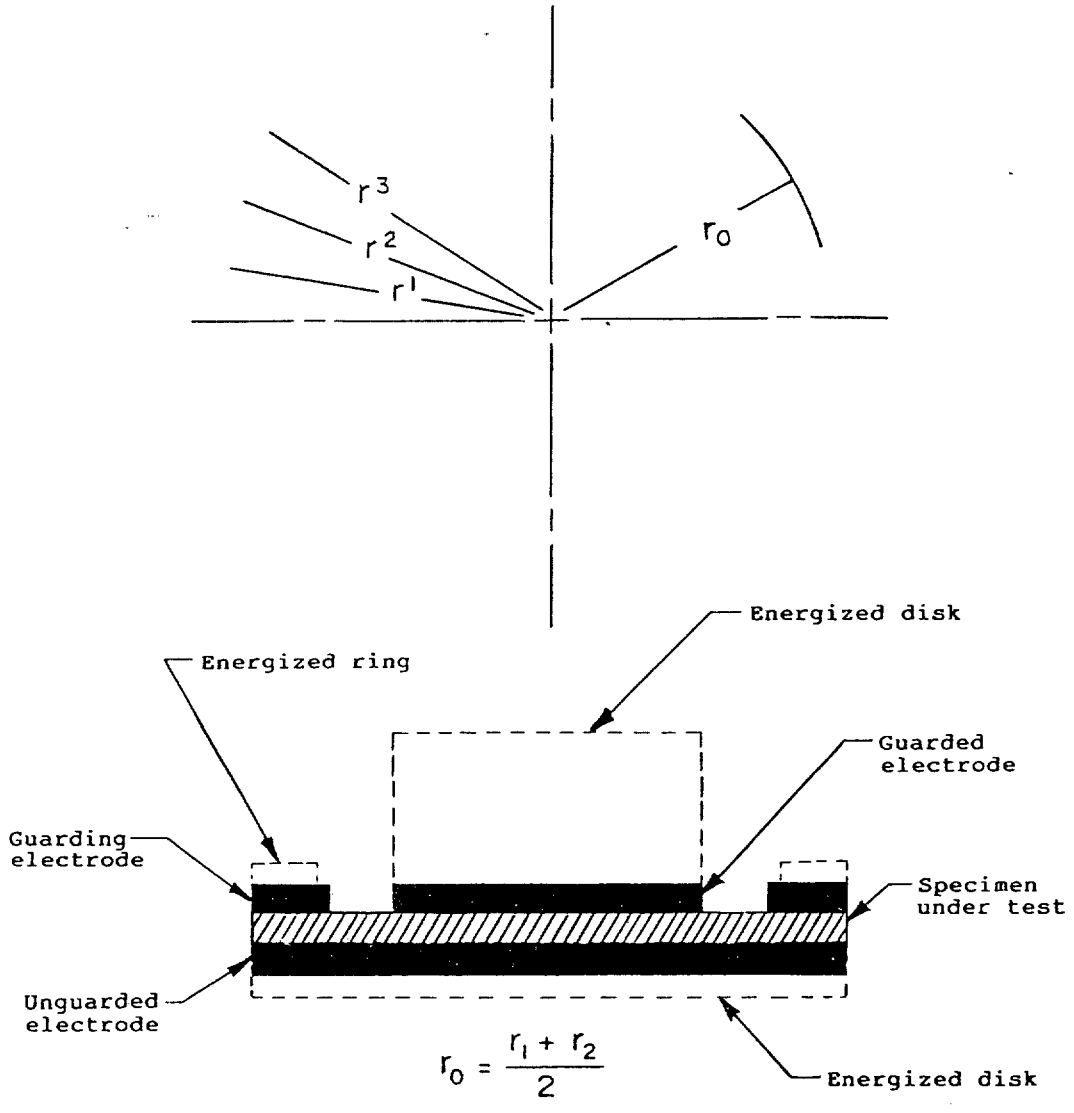


FIGURE 1. Volume resistivity specimen.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-47315(MI)	ADHESIVE, POLYURETHANE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:08Z
Check the source to verify that this is the current version before use.